IB MATHEMATICS SL • UNIT 4.01

C Topic 4: Statistics, Regression & Probability Distributions

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

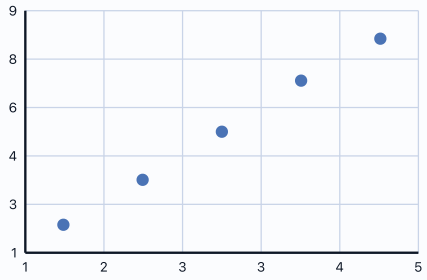
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 B	Q3 D	Q4 A	Q5 A
Q6 C	Q7 B	Q8 A	Q9 C	Q10 D
Q11 B	Q12 A	Q13 C	Q14 B	Q15 C
Q16 C	Q17 D	Q18 C	Q19 B	Q20 B
Q21 B	Q22 A	Q23 C	Q24 B	Q25 D

Topic 4: Statistics, Regression & Probability Distributions Practice Exercises

1 MCQ

If $P(A) = 0.3$ and $P(B) = 0.5$ and A, B are independent, find $P(A \cup B)$.



- **A** 0.65 ✓
- **B** 0.80
- **C** 0.15
- **D** 0.85

WORKED SOLUTION & VERIFICATION

$P(A \cup B) = 0.3 + 0.5 - 0.3 \times 0.5 = 0.65$ **[M1] [A1]**

2 MCQ

The following bar chart shows the number of students who achieved different grades in a mathematics test.\nThe grades are A, B, C, and D. The data is as follows: Grade A: 14 students, Grade B: 22 students, Grade C: 18 students, Grade D: 26 students.\nWhat is the total number of students who took the test?

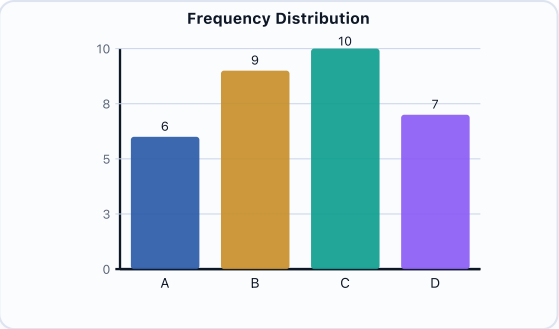
- **A** 70
- **B** 80 ✓
- **C** 78
- **D** 60

WORKED SOLUTION & VERIFICATION

To find the total number of students who took the test, we need to sum the number of students who achieved each grade.\nNumber of students with Grade A = 14\nNumber of students with Grade B = 22\nNumber of students with Grade C = 18\nNumber of students with Grade D = 26\nTotal number of students = 14 + 22 + 18 + 26 = 80.\nTherefore, 80 students took the test. **[M1] [A1]**

3 MCQ

The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students surveyed?



A 50

B 60

C 70

D **80** ✓

WORKED SOLUTION & VERIFICATION

The bar chart shows the following counts for each sport:
Sport 1: 14 students
Sport 2: 22 students
Sport 3: 18 students
Sport 4: 26 students
To find the total number of students surveyed, we sum the number of students for each sport: $14 + 22 + 18 + 26$.
$14 + 22 = 36$
$18 + 26 = 44$
$36 + 44 = 80$.
Therefore, the total number of students surveyed is 80. **[M1] [A1]**

4 MCQ

The histogram below shows the number of hours 100 students spent studying for an exam. The number of students who studied for between 0 and 5 hours is represented by a bar with a height of 30. Determine the number of students who studied for between 5 and 10 hours.

A **20** ✓

B 25

C 30

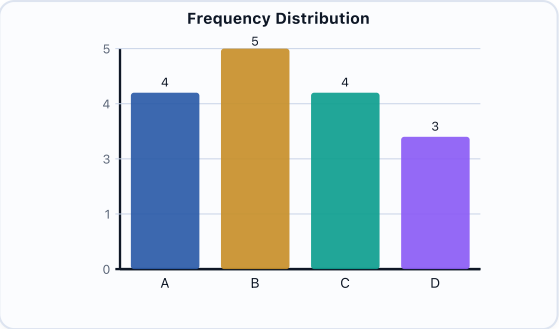
D 35

WORKED SOLUTION & VERIFICATION

The histogram displays the distribution of study hours for 100 students. The total number of students is 100. The question states that the number of students who studied for between 0 and 5 hours is 30. The bar for the '0-5 hours' interval has a height of 30, representing 30 students. The bars for the '5-10 hours' and '10-15 hours' intervals have heights of 20 and 15 respectively, representing 20 and 15 students. Therefore, the number of students who studied for between 5 and 10 hours is 20. **[M1] [A1]**

5 MCQ

The bar chart below shows the number of students who achieved a certain score on a test. What is the total number of students who scored 6 or 8?



- **A** 30 ✓
- **B** 28
- **C** 26
- **D** 32

WORKED SOLUTION & VERIFICATION

From the bar chart, the number of students who scored 6 is 18 and the number of students who scored 8 is 12. The total number of students who scored 6 or 8 is the sum of these two values: $18 + 12 = 30$. **[M1] [A1]**

6 MCQ

The following two-way table shows the results of a survey on favourite sports among a group of students. Calculate the total number of students surveyed.

- **A** 50
- **B** 45
- **C** 60 ✓
- **D** 55

WORKED SOLUTION & VERIFICATION

The total number of students surveyed is the sum of all entries in the two-way table. $\text{Total} = 15 + 10 + 20 + 15 = 60$. **[M1] [A1]**

7 MCQ

Find the mean of the data set: $20, 13, 12, 12, 8$.

- **A** 10
- **B** 13 ✓
- **C** 8
- **D** 11

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $\text{Mean} = \dfrac{20 + 13 + 12 + 12 + 8}{5} = 13$. **[M1] [A1]**

8 MCQ

Find the mean of the data set: $10, 1, 5, 14, 3, 7, 0, 8$.

- **A** 6 ✓
- **B** 4
- **C** 3
- **D** 1

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{10+1+5+14+3+7+0+8}{8} = 6$. **[M1] [A1]**

9 MCQ

Find the mean of the data set: $31, 3, 13, 4, 5, 4$.

- **A** 6
- **B** 5
- **C** 10 ✓
- **D** 7

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{31+3+13+4+5+4}{6} = 10$. **[M1] [A1]**

10 MCQ

Find the mean of the data set: $-2, 6, 7, 8, 11, 6$.

- **A** 4
- **B** 3
- **C** 1
- **D** 6 ✓

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{-2+6+7+8+11+6}{6} = 6$. **[M1] [A1]**

11 MCQ

Calculate the mean of the data set: $17, 20, 19, 25, 14$.

- **A** 16
- **B** 19 ✓
- **C** 14
- **D** 15

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{17+20+19+25+14}{5} = 19$. **[M1] [A1]**

12 MCQ

Find the mean of the data set: $12, -10, 1, 7, 14, 3, 9, 12$.

- **A** 6 ✓
- **B** 4
- **C** 1
- **D** 3

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{12 - 10 + 1 + 7 + 14 + 3 + 9 + 12}{8} = 6$. **[M1] [A1]**

13 MCQ

A bag contains 4 red balls, 3 green balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

- **A** $\frac{8}{11}$
- **B** $\frac{2}{11}$
- **C** $\frac{3}{11}$ ✓
- **D** $\frac{4}{11}$

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $P(\text{green}) = \frac{3}{11} = \frac{3}{11}$. **[M1] [A1]**

14 MCQ

A bag contains 4 green balls, 3 blue balls, and 2 red balls. A ball is drawn at random. Find $P(\text{blue})$.

- **A** $\frac{4}{9}$
- **B** $\frac{1}{3}$ ✓
- **C** $\frac{2}{9}$
- **D** $\frac{2}{3}$

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $P(\text{blue}) = \frac{3}{9} = \frac{1}{3}$. **[M1] [A1]**

15 MCQ

A bag contains 8 green balls, 6 blue balls, and 6 red balls. A ball is drawn at random. Find $P(\text{red})$.

- **A** $\frac{7}{20}$
- **B** $\frac{2}{5}$
- **C** $\frac{3}{10}$ ✓
- **D** $\frac{7}{10}$

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $P(\text{red}) = \frac{6}{20} = \frac{3}{10}$. **[M1] [A1]**

16 MCQ

Find the mean of the data set: $2, 14, -3, 0, 7$.

A 1

B 2

C 4 ✓

D 0

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{2+14-3+0+7}{5} = 4$. **[M1] [A1]**

17 MCQ

A bag contains 3 green balls, 7 red balls, and 4 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

A $\frac{2}{7}$

B $\frac{1}{2}$

C $\frac{11}{14}$

D $\frac{3}{14}$ ✓

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $P(\text{green}) = \frac{3}{14} = \frac{3}{14}$. **[M1] [A1]**

18 MCQ

Calculate the mean of the data set: $11, 17, 22, 27, 12, 25$.

A 14

B 16

C 19 ✓

D 15

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{11+17+22+27+12+25}{6} = 19$. **[M1] [A1]**

19 MCQ

A bag contains 4 red balls, 2 blue balls, and 2 yellow balls. A ball is drawn at random. Find $P(\text{blue})$.

A $\frac{1}{2}$

B $\frac{1}{4}$ ✓

C $\frac{3}{8}$

D $\frac{3}{4}$

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $P(\text{blue}) = \frac{2}{8} = \frac{1}{4}$. **[M1] [A1]**

20 MCQ

A bag contains 4 yellow balls, 6 blue balls, and 3 green balls. A ball is drawn at random. Find $P(\text{green})$.

- A $\frac{6}{13}$
- B $\frac{3}{13}$ ✓
- C $\frac{4}{13}$
- D $\frac{10}{13}$

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $P(\text{green}) = \frac{3}{13} = \frac{3}{13}$. **[M1] [A1]**

21 MCQ

A bag contains 5 green balls, 5 red balls, and 3 yellow balls. A ball is drawn at random. Find $P(\text{yellow})$.

- A $\frac{4}{13}$
- B $\frac{3}{13}$ ✓
- C $\frac{5}{13}$
- D $\frac{10}{13}$

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $P(\text{yellow}) = \frac{3}{13} = \frac{3}{13}$. **[M1] [A1]**

22 MCQ

Find the mean of the data set: $9, 9, -3, 4, 1, 11, 12, -3$.

- A 5 ✓
- B 3
- C 1
- D 2

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{9+9-3+4+1+11+12-3}{8} = 5$. **[M1] [A1]**

23 MCQ

Compute the mean of the data set: $12, 9, 13, 12, 29$.

- A 10
- B 12
- C 15 ✓
- D 11

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{12+9+13+12+29}{5} = 15$. **[M1] [A1]**

24 MCQ

A bag contains 4 green balls, 4 red balls, and 5 yellow balls. A ball is drawn at random. Find $P(\text{green})$.

A $\frac{5}{13}$

B $\frac{4}{13}$ ✓

C $\frac{3}{13}$

D $\frac{9}{13}$

WORKED SOLUTION & VERIFICATION

Working through this statistics step: $P(\text{green}) = \frac{4}{13} = \frac{4}{13}$. **[M1] [A1]**

25 MCQ

Calculate the mean of the data set: $22, 26, 15, 26, 16, 20, 13, 14$.

A 14

B 15

C 16

D **19** ✓

WORKED SOLUTION & VERIFICATION

Working through this statistics step: Mean $= \frac{22+26+15+26+16+20+13+14}{8} = 19$. **[M1] [A1]**